

Binomial Asset Pricing Model

Volume 1, Chapter 1

§1. Binomial Asset Pricing Model

§1.1. Notation

- **Time 0:** Very start of the period. **Time 1:** End of the first period.
- S_0 : Stock price at time 0. $S_1(H)$ is price at time 1 if coin flip is heads, $S_1(T)$ is price at time 1 if coin flip is tails.
- p is probability of heads, $q = 1 - p$ is probability of tails.
- r is the interest rate, where 1 dollar invested at time zero yields $1 + r$ at time one.
- K is the strike price for European call options.

§1.2. One-Period Binomial Model

Main Idea: Consider the following diagram.

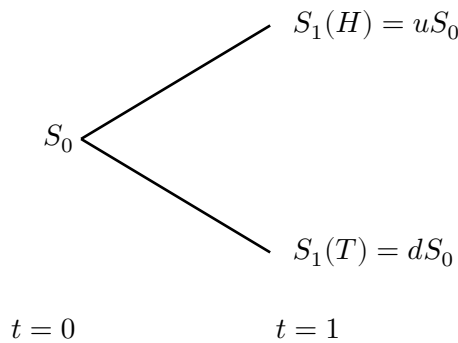


Figure 1.2.1: One-period binomial model

We know the outcome of the coin toss at time 1, but not time 0. Define

$$u = \frac{S_1(H)}{S_0}, \quad d = \frac{S_1(T)}{S_0}.$$

with $d < u$.

Definition 1.2.1 (Arbitrage).

A trading strategy that begins with no money, has zero probability of losing money, and positive probability of making money.

Remark 1.2.1.

We assume that there is no arbitrage in our analysis. Otherwise, we get contradictions because we can basically get infinite wealth generation.

Remark 1.2.2.

To prevent arbitrage, we must have $0 < d < 1 + r < u$.

Otherwise, if $d \geq 1 + r$, borrow money and buy the stock with the borrowed money, and the stock will always earn enough to pay off the borrowed money. If $u \leq 1 + r$, short the stock and use the proceeds in the money market, which will earn enough interest to buy back the stock at time 1. The converse is true.

Stock prices obviously do not move like this, but given a sufficient number of periods, this helps approximate continuous-time models.

Definition 1.2.2 (European Call Option).

A European call option gives its owner the right, but not the obligation, to buy one share of the stock at time one for strike price K at a certain point in time.

We consider the case where $S_1(T) < K < S_1(H)$. Observe that if we flip tails, the option expires worthless, but if we flip heads, the option has a payoff of $S_1(H) - K$.

Definition 1.2.3 (Arbitrage Pricing Theory).

Pricing options by replicating the option by trading in the stock and money markets.

Remark 1.2.3.

If we can get the same returns using the stock and money markets as we get from the option, then they should be valued at the same price.

Example 1.2.1.

Consider the following:

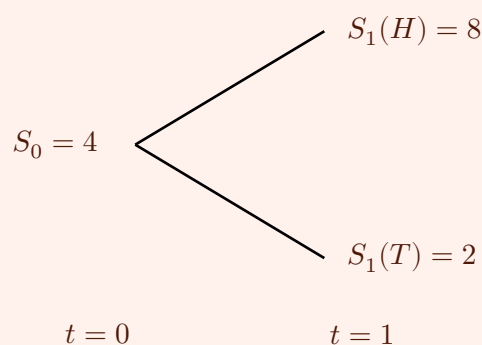


Figure 1.2.2: Example one-period binomial model

With $r = \frac{1}{4}$ and $K = 5$. Observe that if we have an initial wealth of $X_0 = 1.20$, borrow 0.80 to spend 2.00 on $\frac{1}{2}$ shares at time zero, then

$$X_1(H) = \frac{1}{2}S_1(H) - (1 + r)(0.80) = 3$$

and

$$X_1(T) = \frac{1}{2}S_1(T) - (1 + r)(0.80) = 0$$

which match the payoffs of the option! We have replicated the European call option, meaning that the option **must be worth the initial wealth** of 1.20.

If the option was worth more than 1.20, we could short the option and buy the replicating portfolio (i.e. buy $\frac{1}{2}$ shares and borrow 0.80). If the option was worth less than 1.20, we could buy the option and short the replicating portfolio (i.e. short sell $\frac{1}{2}$ shares and invest 0.80 in the money market). Both cases result in an **arbitrage opportunity**.

There are a few assumptions made:

- We can buy fractional shares.
- Interest rate for investing is the interest rate for borrowing.
- Purchase price of stock is the same as its selling price.
- Stock prices can be only one of two values in the next period.

Definition 1.2.4 (Derivative Security).

A derivative security is a security that pays, at time 1, $V_1(H)$ if the coin toss is heads and $V_1(T)$ if the coin toss is tails

To get the price of a derivative security at time 0, we replicate the security. We can derive a formula for this. Let X_0 be the value of the replicating portfolio, and hence, the **value of the option**. Let Δ_0 be the number of shares of stock we buy at time 0. The value of the replicating portfolio at time 1 is

$$X_1 = \underbrace{\Delta_0 S_1}_{\text{Position in stocks}} + \underbrace{(1+r)(X_0 - \Delta_0 S_0)}_{\text{Amount due on borrowed}} = \Delta_0(S_1 - (1+r)S_0) + (1+r)X_0$$

We want to find X_0 and Δ_0 so that $X_1(T) = V_1(T)$ and $X_1(H) = V_1(H)$.

Define $\tilde{p}$ and $\tilde{q} = 1 - \tilde{p}$ to be the values satisfying

$$S_0 = \frac{1}{1+r} [\tilde{p}S_1(H) + \tilde{q}S_1(T)]$$

This is similar to calculating the expected value for S_0 . Equating $X_1(H) = V_1(H)$ and $X_1(T) = V_1(T)$, we have

$$\begin{aligned} X_0 + \Delta_0 \left(\frac{1}{1+r} S_1(H) - S_0 \right) &= \frac{1}{1+r} V_1(H) \\ X_0 + \Delta_0 \left(\frac{1}{1+r} S_1(T) - S_0 \right) &= \frac{1}{1+r} V_1(T) \end{aligned} \tag{*}$$

We can add these equations together, multiplied by $\tilde{p}$ and $\tilde{q}$, to get

$$X_0 + \Delta_0 \left(\frac{1}{1+r} [\tilde{p}S_1(H) + \tilde{q}S_1(T)] - S_0 \right) = \frac{1}{1+r} [\tilde{p}V_1(H) + \tilde{q}V_1(T)].$$

Because of how we defined $\tilde{p}$ and $\tilde{q}$, we get

$$X_0 = X_0 + \Delta_0(S_0 - S_0) = \frac{1}{1+r} [\tilde{p}V_1(H) + \tilde{q}V_1(T)].$$

To solve for $\tilde{p}$, we have

$$S_0 = \frac{1}{1+r}(\tilde{p}uS_0 + (1-\tilde{p})dS_0)$$

$$1+r = \tilde{p}u + (1-\tilde{p})d$$

$$\tilde{p} = \frac{1+r-d}{u-d}.$$

Hence, we obtain $\tilde{q} = \frac{u-1-r}{u-d}$. To solve for Δ_0 , we subtract the equalities in * to obtain

$$\Delta_0 = \frac{v_1(H) - V_1(T)}{S_1(H) - S_1(T)}$$

Therefore, we should value the security at

$$V_0 = \frac{1}{1+r}[\tilde{p}V_1(H) + \tilde{q}V_1(T)]$$

Definition 1.2.5 (Risk Neutral Probabilities).

We call $\tilde{p}$ and $\tilde{q}$ the risk-neutral probabilities.

Remark 1.2.4.

$\tilde{p}$ and $\tilde{q}$ are indeed probabilities as they sum to 1 and are non-negative. They represent the probabilities of the coin flip so that the **expected** growth rate of the stock is equal to r . In fact, because investors compensate for risk, if p and q represent the actual probabilities of the coin toss,

$$S_0 < \frac{1}{1+r}[pS_1(H) + qS_1(T)]$$

$$\frac{1}{1+r}[\tilde{p}S_1(H) + \tilde{q}S_1(T)] < \frac{1}{1+r}[pS_1(H) + qS_1(T)]$$

Remark 1.2.5.

We do not worry about the actual probabilities for valuation! We have constructed a replicating portfolio for the option, so the probabilities of up and down are irrelevant. Our replicating portfolio must earn the risk-free return.

§1.3. Multiperiod Binomial Model

§2. Binomial Asset Pricing Model

§2.1. One-Period Binomial Model